

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Process Design Philosophy
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PSE-PHY-001
Document Revision	R01
Document Status	Issued for Review
Originator	Akinwale Biobaku
Security Classification	Restricted
Issue Date	14-Jul-22

Process Design Philosophy

R01	14.07.22	Issued for Review	B. Akinwale	M. Ekpunobi	K. Yusuff	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	14-07-22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PSE-PHY-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-GEN-PHY-001	Design & Operating Philosophy
EA-BFPCL-GPMC-GGPL-FEED-PSE-PHY-002	Process Control & Safeguarding Philosophy
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report

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Abbreviations

Abbreviation	Meaning
API	American Petroleum Institute
BFPCL	Brass Fertilizer and Petrochemical Company Limited
BOD	Basis Of Design
DEP	Design and Engineering Practice
DPR (now NUPRC)	Department for Petroleum Resources
FEED	Front End Engineering Design
GPP	Gas Processing Plant
MMSCFD	Metric Million Standard Cubic Feet Per Day
MTPD	Metric Tonnes Per Day
NUPRC (formerly DPR)	Nigerian Upstream Petroleum Regulatory Commission
OML	Oil Mining License
PIG	Pipeline Inspection Gauge
PMC	Project Management Consultant
RTU	Remote Telemetry Unit
SPDC	Shell Petroleum Development Company

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

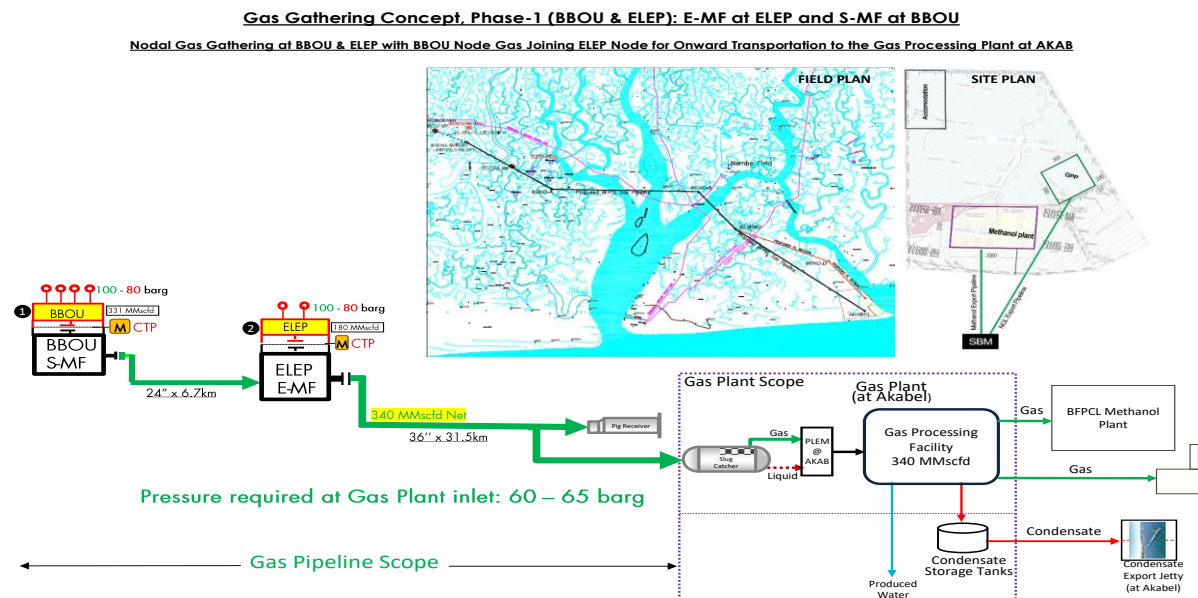


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this document is to recommend minimum requirements and guidelines for the design and installation of process piping and equipment systems for the gas gathering Project. The recommendations practices presented are based on years of experience in developing oil and gas leases.

2. Regulations, Codes And Standards

Document Number	Document Title
Petroleum Act, CAP P10, Laws of the Federation of Nigeria, LFN 2004, amended.	This is the principal statute regulating the oil industry. The following regulation relevant to this project is issued pursuant to the Act.
Mineral Oil (Safety) Regulations 1963, as amended.	These regulations, in so far as they apply to pipelines, require compliance with relevant IP, API and ASME codes and standards.
Oil Pipelines Act, 1956. CAP. 145 (Amended 1965, CAP. 338, LFN 1990)	Provides for licenses to be granted for the establishment and maintenance of pipelines incidental and supplementary to oilfields and oil mining, and for purposes ancillary to such pipelines.
Oil and Gas Pipelines Regulations, 1995	Details the regulations for the design, construction and maintenance of oil and gas pipelines.
Environmental Impact Assessment Act, CAP E12, LFN 2004.	Makes the conduct of an EIA mandatory prior to the development of any project or activity likely to have a significant effect on the environment
Federal Environment Protection Agency Act, CAP F13, LFN 2004.	Provides for the protection of the environment and sustainable development of Nigeria's natural resources.
Environmental Guidelines and Standards For The Petroleum Industry In Nigeria, 1991.	DPR (NUPRC) issued document providing guidelines on pollution abatement and relevant legislation.
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA/SPDC-BFPC UGS-GEP/SVY-RPT-001	Pipeline Route Selection Study
EA/SPDC-BFPC UGS-GEP/PRC-RPT-001- 3	Process Simulation Study – HYSYS, PIPESIM, & OLGA
EA/SPDC-BFPC UGS-GEP/MAT-RPT-001	Pipeline Materials Selection & Corrosion Analysis Study

EA/SPDC-BFPC UGS-GEP/ELE-RPT-001	Manifold Platforms Power Supply Options
EA/SPDC-BFPC UGS-GEP/MEC-RPT-001	Manifold Platforms Concept & Integration with SPDC
EA/SPDC-BFPC UGS-GEP/PRC-RPT-002	Gas Processing Technology Option
6064/PLRS/001 a-PRP	Onshore Pipeline Route Survey for BFCL; Topographical Surveys
6064/PLRS/001 PRP	Onshore Pipeline Route Survey for BFCL; Soil Investigation Surveys
40km Gas Supply Pipeline Project ESIA	Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited - Draft Report November 2020.

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Pipe and Pipeline Sizing

3.1. Design Basis

The design pressure of each component in the piping system shall not be less than the following:

1. When the pressure is not limited by a relieving device: the maximum pressure that can be attained in service. Allowances for occasional pressure variations as permitted by Section 302.2.4 "Allowances for Pressure and Temperature Variations" of ASME B31.3 are not permitted except on a case-by-case basis with BFPCL approval. Overpressure associated with the operation of a relief device (i.e., 10% overpressure) as permitted by Section 322.6.3 "Pressure Relieving Devices" of ASME B31.3 is acceptable.
2. When the pressure is limited by a relieving device: the set pressure of the pressure-relieving device.

A maximum and minimum design temperature for each component in a piping system should be established. These temperatures shall be based on the maximum and the minimum temperatures to which the piping shall be subjected to, including start-up, ambient effects, depressurizing for maintenance, and other non-normal operations.

The design temperature of the system shall be as follows:

1. The maximum operating temperature plus an incremental temperature as a safety factor that also addresses solar radiation and other ambient effects.
2. The minimum operating temperature, minus an incremental temperature as a safety factor that covers effects of autorefrigeration.
3. The design temperature of each component in a piping system shall be the highest (or lowest for low temperature systems) to which the piping system shall be subjected.

3.2. Pipe Grades.

The two most commonly used types of pipe are ASTM A106, Grade B, and API 5L, Grade B. Seamless pipe will be used on this Project due to its consistent quality. ASTM A106 is only manufactured in seamless while API 5L is available in seamless, electric resistance welded (ERW) and submerged arc welded (SAW). When use of Grade B requires excessive wall thickness, higher strength pipe such as API 5L, Grade X52, may be required; however, special welding procedures and close supervision are necessary when using API 5L, Grade X46, or higher.

Many of the grades of pipe listed in ANSI B31.3 are suitable for the gas gathering piping systems. The following types or grades of pipe shall not be used in any part of the gas gathering system.

- All grades of ASTM A120.
- Furnace lap weld and furnace butt weld.
- Fusion weld per ASTM A134 and A139.
- Spiral weld, except API 5L spiral weld.

3.3. Pipe Nominal Sizes

The following pipe sizes shall be used:

Pipe sizes of NPS 3/8, 1 1/4, 2 1/2, 3 1/2, 4 1/2, 5, 7, and 9 shall not be used unless required for mating to equipment. The Equipment Supplier is required to provide mating flanges and any associated hardware for nonstandard flanges in the event that a connection to skid equipment must connect to a nonstandard flange.

- Due to their vulnerability to damage and their limited mechanical strength, sizes NPS 1/2 and NPS 3/4 should not be used for long run piping.
- Nominal pipe size in pipe tracks shall not be less than NPS 2;
- Nominal pipe size in pipe racks shall not be less than NPS 1 1/2;
- Pipe sizes between those given in the piping classes shall not be used.

Vent, drain and equipment auxiliary piping shall be NPS 3/4 minimum, except orifice taps, which shall be NPS 1/2.

Instrument tubing and heat tracing lines shall be a minimum of NPS 1/2 pipe or 10 mm (3/8 in.) OD.

Since small bore branches (≤ DN 1 1/2) to large bore piping are relatively susceptible to failure, the following points shall be incorporated in the piping design:

- Minimise the number of small-bore branches to piping.
- All small-bore nozzle welding shall be performed by Gas Tungsten Arc Welding (GTAW) in the prefabrication yard or shop.
- Small bore piping shall be shown in full detail on the isometric drawings.
- Branches shall not be in removable spools.
- Branches shall not be in high stress areas.
- Branches shall be avoided downstream of high-capacity gas pressure reducing systems such as compressor recycle systems, steam desuperheaters, high-rate depressuring valves and safety relief valves (acoustic fatigue). If this is not possible, branches shall be located well away from these sources of vibration.
- In branches with flanged valves, branch fittings with flanged outlets shall be used wherever possible, to reduce the number of welds.

3.4. Pipe Sizing

The following should be considered in determining a suitable pipe size:

- The available pressure drop.
- The fact that pressure surges may occur in the piping system.
- The fact that erosion may occur in the piping system.
- The fact that the piping system may be subjected to vibrations.
- The fact that settlement of solids if the fluid is a slurry may occur.
- Type of flow pattern of two-phase flow. An intermittent flow pattern shall be avoided.
- The allowable temperature drop if the fluid is highly viscous.
- An economic pipe diameter considering the capital expenditure and operating expenditure of the pumps, compressors and the piping system.

The above considerations shall be taken into account both for the design capacity and for conditions such as starting up, shutting down and regeneration.

In general, sizing criteria shall follow the guidelines in API RP 14E. The following velocities should be used for preliminary pipe sizing for interconnecting piping:

Process	Min Velocity	Max Velocity
water and other liquids	0.5	5 m/s;
air and other gases	10	20 m/s;
saturated steam (dry)	15	30 m/s;
superheated steam	30	60 m/s;
vacuum pipes	10	100 m/s.

The above figures are not applicable to sludge, slurry, erosive fluids or two-phase flow. Flowlines, production manifolds, process headers and other lines transporting gas and liquid in two-phase flow should be sized primarily on the basis of flow velocity ranging less than erosional velocity. The guidelines of API 14E for determining erosional velocity shall be used in determining limiting velocity in the pipes or pipelines.

For sizing of flare and vent systems, line sizing shall generally be in accordance with API RP 520 PT I, API RP 520 PT II, API RP 521 and DEP 80.45.10.10-Gen.

3.4.1. Pressure drop calculations

3.4.1.1. General

The formulae for pressure drop calculations given in this section are applicable to fluids whose density and viscosity are constant along the length of the whole pipe, which is normally the case for liquids. Pressure drop due to piping components other than straight pipe shall be expressed for calculation purposes as equivalent lengths (L_e) and added to the length of the straight pipe in the system, thus giving the total length (L) to be used in the pressure drop calculations.

For valves and fittings, L_e is given in the table below:

Type of valve/fitting			*Le (m)
Valves (fully open)	Ball valve	Reduced bore 40 mm and smaller	65 D
		Reduced bore 50 mm and larger	45 D
	Gate valve	Standard bore	13 D
		Reduced bore 40 mm and smaller	65 D
	Globe valve	Straight pattern	340 D
		Y pattern	160 D
		Angle pattern	145 D
Check valve	Swing type	135 D	
	Ball or piston type, 40 mm and smaller	340 D	
Plug valve	Regular pattern	45 D	
Butterfly valve	150 mm and larger	20 D	
Fittings	Tee-equal	Flow straight through	20 D
		Flow through side outlet	65 D
	Elbow	90°, R = 1 1/2 D	20 D
		45°, R = 1 1/2 D	16 D
	Bend	90°, R = 4 D	14 D
		90°, R = 5 D	16 D
		180°, R = 4 D	25 D
180°, R = 5 D		28 D	
Miscel- laneous	Strainer	Pump suction Y-type and bucket type	250 D
	Nozzle	Product outlet nozzle vessel/tank	32 D
Product inlet nozzle vessel/tank		64 D	

The pressure drop across control valves shall be determined in consultation with parties responsible for instrumentation. To obtain a reasonable degree of control, the pressure drop across the control valve (at 80% open position) shall be at least 1 bar. However, if the pressure drop is too high it should be considered to install small bore piping to reduce the pressure drop across the valve.

The pressure drop of in-line instruments, such as vortex or Coriolis flow meters, shall be determined from the Manufacturer's data.

For equipment, the pressure drop shall be taken from the data obtained from the equipment Manufacturer.

4. Pressure Relief Devices

4.1. Design

Pressure relief device designs shall follow practices recommended in API RP 520 PT I, API RP 520 PT II and API RP 521 with exceptions and additions noted herein. Pressure relief requirements for all equipment and piping systems shall be per ASME Code Section VIII D1 and D2 and shall be sized in accordance with API RP 520 PT I and the Shell DEP, whichever is more stringent.

4.2. Types of Devices

Acceptable types of PR valves are: conventional, balanced bellows, or pilot-operated. The use of pilot operated or balanced bellows PR valves shall be approved by the Owner's Engineer. Soft seated PR valves shall be considered as an alternative to rupture disc/PR valve combinations for services operating below 450F (232C) when control of fugitive emissions is required to meet environmental regulations. The use of soft seated PR valves shall be approved by the Owner's Engineer.

Rupture discs (RD) may be used in conjunction with or as a substitute for PR valves only with the approval of the Owner's Engineer. Acceptable types of RDs are:

- i. Pre-scored tension loaded conventional (forward acting) for both liquid and gas/two-phase service.
- ii. Pre-scored (cross-scored) reverse buckling for gas/two-phase only service.
- iii. Pre-scored (semi-circular) reverse buckling for gas/two-phase or liquid service. Pre-scored (semi-circular) may not be used under PRVs due to possible fragmentation unless a special holder is used.

The manufacturing range for all RDs shall be the lowest available from the manufacturer, preferably zero. The full amount of the manufacturing range shall be applied below the design pressure of the equipment. Use of other types of RDs with larger manufacturing ranges shall be approved by the Owner's Engineer.

Buckling pin valves (BPV) may be used in conjunction with, or as a substitute for, PR valves only with the approval of the Owner's Engineer. Conventional or balanced BPVs are acceptable. If the BPV contains an internal check valve, it shall be used in clean service only. BPV are recognized by ASME Code Case 2091.

4.3. Valve Sizing

The minimum size of PR valves shall be as follows:

- a) 1 in. NPS (25 mm) inlet size with flanged inlet and outlet connections.
- b) 3/4 in. NPS (20 mm) when used only for protection against liquid thermal expansion. This size may be used with threaded inlet and outlet connections.

Smaller sizes shall be approved by Owner's Engineer. Vessels fabricated from pipe less than 24 in. NPS (600 mm), which are not ASME Code stamped, do not require PR devices for protection against fire. Unless otherwise specified, pressure relief valves with Q, R, and T size orifices shall not be used where the temperature exceeds 350°F (177°C) and the molecular weight is less than 10 without Owner Engineer's approval. Pressure relief valves with orifice sizes larger than T or with inlet flanges larger than NPS 8 in. (200 mm) shall not be used without Owner Engineer's approval.

5. Instrument Take-Off Piping

The minimum flow rates for Purge Systems at process takeoff connections shall be as shown below:

Instrument Service	Minimum Velocity	Purge
Liquids	0.25 in./sec, 6 mm/sec	Liquid Purge
Gas Solids	24 in./sec, 600 mm/sec	Gas Purge
Cracking	3 ft/sec, 0.9 m/sec	Gas Purge
Coking	6 ft/sec, 1.8 m/sec	Gas Purge

Note (1): Higher velocities may be required to prevent sensor burnout. Each installation shall be approved by Owner's Engineer.

6. Pigging System

6.1. Design Code

The entire pig trap system should be designed, constructed and tested according to the same code as the pipeline. However, various options of design code break between ANSI B31.8/B31.4 and ANSI B31.3 may be used.

Wall thickness transitions between B31.3 and Pipeline Codes shall meet the welding

6.2. Design Factor

The design factor shall be as stated on the requisition and should be applied to all piping of the pig trap system. This assumes that the traps are designed to B31.4/8 and is not applicable where ANSI B31.3 is the chosen code.

6.3. Design Pressure

The design pressure of the pig trap system shall not be less than that of the pipeline. Pipeline design pressure and fitting class rating shall be specified by the Principal.

6.4. Design Temperature

The maximum design temperature shall not be less than the maximum temperature which the pig trap system could attain or to which it could be exposed during operation, start-up or shutdown.

The minimum design temperature shall be based on minimum ambient temperature and on the conditions (e.g. blowdown) which could occur during operations.

Minimum and maximum ambient temperatures shall be specified by the BFPCL.

6.5. Design Velocities

Suggested maximum velocities for the purpose of pigging piping system are:

For piping in intermittent service:

- In case of liquid - 8 m/s.
- In case of gas - 40 m/s.

For piping in continuous service:

- In case of liquid - 4 m/s.
- In case of gas - 20 m/s.

Based on these velocities and the assumption that parts of the pig trap are in intermittent service, piping diameters are suggested in the Shell DEP 31.40.10.13-Gen. For every design, however, it should be checked that the velocities do not exceed designated maxima and that piping pressure drops are not excessive.

6.6. Test Pressure

The pig trap system may be hydrostatically tested either together with, or separately from, the pipeline. In either case the test pressure shall not be less than that of the adjacent pipeline section.